INCH POUND
MIL-DTL-22775E
16 May 2023
SUPERSEDING
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06 January 1984

DETAIL SPECIFICATION

CLOTH, BUNTING, ACRYLIC

This specification is approved for use by Navy Clothing and Textile Research Facility.

1. SCOPE

1.1 Scope. This specification covers the requirements for an acrylic bunting cloth.

1.2 Classification.

1.2.1. Classes. The cloth will be of one type only, in the following classes (see 6.2).

Class 1 – White 3032
Class 2 – Black 3227
Class 3 – Blue 3341
Class 4 – Green 3429
Class 5 – Yellow 3510
Class 6 – Red 3803

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support, 700 Robbins Street, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database https://assist.dla.mil.

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2.1 Government documents.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

FEDERAL TRADE COMMISSION (FTC)

16 CFR Part 303 - Rules and Regulations under the Textile Fiber Product Identification Act

(Copies are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Evaluation Procedures:
AATCC EP1 - Gray Scale for Color Change
AATCC EP2 - Gray Scale for Staining
AATCC EP8 - 9 Step Chromatic Transference Scale Rating
AATCC EP9 - Visual Assessment of Color Difference of Textiles

Test Methods:
AATCC TM8 - Colorfastness to Crocking: Crockmeter
AATCC TM16.2 - Colorfastness to Light: Carbon Arc
AATCC TM16.3 - Colorfastness to Light: Xenon
AATCC TM20A - Fiber Analysis: Quantitative
AATCC TM61 - Colorfastness to Laundering: Accelerated
AATCC TM81 - pH of Water-Extract from Wet Processed Textiles
AATCC TM94 - Finishes in Textiles: Identification
AATCC TM106 - Colorfastness to Water: Sea
AATCC TM135 - Dimensional Changes of Fabrics after Home Laundering
AATCC TM169 - Weather Resistance of Textiles: Xenon Lamp Exposure

(Copies are available online at https://www.aatcc.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Fall

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ASTM D1776/D1776M - Standard Practice for Conditioning and Testing Textiles
ASTM D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight of Fabric)
ASTM D3990 - Standard Terminology Related to Fabric Defects
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies are available online at https://www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified, the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.3 and 6.2).

3.2 Material.

3.2.1 Fiber. The fiber shall be first quality, 3-denier, semi-dull acrylic staple with a minimum staple length of 2-inches, see 4.4.1.

3.2.2 Yarn. The acrylic fiber shall be evenly spun and twisted into a two-ply yarn for both the warp and filling, see 4.4.1.

3.2.3 Weave. The cloth shall be a bunting plain weave, see 4.4.1.

3.2.4 Color. The color of the finished cloth shall be as specified below in the following classes and shall match its standard sample, see 3.2.6, 4.4.2.

Class 1 – White 3032
Class 2 – Black 3227
Class 3 – Blue 3341
Class 4 – Green 3429
Class 5 – Yellow 3510
Class 6 – Red 3803

3.2.5 Finish. The cloth shall be dyed, finished and heat set equal to or better than the standard sample. The cloth may be finished with an optical brightener/UV stabilizer to achieve shade match requirements in Table I when tested as specified in 4.4.1 and 4.4.2.

3.2.6 Standard sample. The finished cloth shall be equal to the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The finished cloth shall

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match the standard sample for shade, presence of optical brightener, appearance, discoloration, and finish, see 3.2.4 and 4.4.2.

3.2.7 Non-fibrous material. The total sizing, finishing and other non-fibrous material content shall not exceed 2.0 percent when tested as specified in 4.4.1.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.4.1.

TABLE I. Physical requirements

Characteristics	Requirement
Weight, ounces per square yard (minimum)	4.5
Yarns per inch (minimum)	
Warp	28
Filling	29
Breaking strength, pounds	
Initial (minimum)	95
Warp	97
Filling	
After resistance to weathering (minimum)	95
Warp	95
Filling	
Tear strength, pounds	
Initial, (minimum)	6
Warp	6
Filling	
After resistance to weathering (minimum)	5
Warp	5
Filling	
Air permeability ($ft^3/min/ft^2$) minimum	250
Dimensional change, percent (maximum)	
Warp	4.0
Filling	4.0

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TABLE I. Physical requirements - Continued

Characteristics	Requirement
pH	5.0 to 8.5
Colorfastness - All classes (minimum)	
Laundering	
Color Change	3
Staining	3
Sea Water	
Color Change	3
Staining	3
Light (170 kJ@ 420nm or after 40AFU (minimum)	3
Crocking	
Dry	3.0
Wet	3.0

3.5 Width. The minimum acceptable width of the finished cloth, inclusive of selvages, shall be as specified (see 6.2). The selvage shall be 1/4 (± 1/8) inch (6.35 ± 3.18mm) wide. When shuttleless loom is utilized the width of the cloth shall be as specified (see 6.2) and shall be excluding the selvages, see 4.4.6.

3.6 Length. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous pieces, each not less than 40 yards in length. See 4.4.7.

3.6.1 Cloth put-up. Each length of cloth furnished shall be put upon rolls, full width with the face side inward. See 4.4.8.

3.7 Fiber identification label. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act. See 4.4.9.

3.8 Face identification marking. The filling side (back side) of the cloth shall be used as the face side (front side). The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word “Face” at each end of the roll (see 4.4.10.).

3.9 Visual shade matching. The cloth in each roll of a finished lot shall be examined visually for shade match regarding appearance, shade, finish, and shade variation, see 3.2 and 3.3.4. When viewed as specified in 4.4.2, the cloth shall match the standard sample.

3.10 Workmanship. The finished cloth shall be uniform in quality and free from irregular defects that can affect usage and/or durability and the defects specified in ASTM D3990 and Section 4. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required, the inspection shall consist of the examinations and tests specified in 4.4.1 through 4.4.12, See Section 3.

4.2.1 First article sample. Unless otherwise specified in the contract (see 6.2), sample unit shall be (1) continuous five (5) yard length of full width finished cloth.

4.3 Conformance inspection. Conformance testing shall consist of all examinations and tests specified in 4.4.1 through 4.4.12, see Section 3.

4.3.1 Conformance inspection sampling. The sample unit shall be (1) continuous five (5) yard length of full width fabric per roll of finished cloth, except for the shade evaluation cited under 4.5.1 shall not apply. The number of samples per lot size shall be a specified below.

Lot Size (yards)	Sample Size (rolls)
3200 or less	8
3201 up to and including 10,000	13
10, 001 and over	20

4.3.1.1 In-process inspection. Visual and dimensional examinations of the shirt and its components or optional items shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with requirements specified in Section 3. Materials and components, which can be classified, as a defect in accordance with ASTM D3990 for woven cloth and Section 4 shall be removed from production.

4.4 Inspection methods.

4.4.1 Finished cloth tests. The cloth shall be tested as specified in Table II. The finished cloth shall be conditioned as directed in ASTM D1776/D1776M. All test reports shall contain the individual values utilized in expressing the results.

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TABLE II. Finished cloth tests

Characteristics	Requirement	Verification
Fiber analysis	3.2.1	AATCC 20A
Yarn ply	3.2.2	Visual 1/
Weave	3.2.3	Visual 1/
Finish	3.2.5	AATCC TM94
Non-Fibrous	3.2.8	AATCC 20A
Weight	3.4	ASTM D3776/D3776M, Option C
Yarn per inch	3.4	ASTM D3775
Breaking strength, Initial	3.4	ASTM D5034
Breaking strength, after resistance to weathering	3.4	ASTM D5034 and AATCC 169
Tear strength, initial	3.4	ASTM D1424
Tear strength, after resistance to weathering	3.4	ASTM D1424 and AATCC 169
Air Permeability	3.4	ASTM D737
Dimensional change (1 cycles)	3.4	AATCC 135 (3) V Aiii, 2/
pH	3.4	AATCC 81
Colorfastness to – All Classes:		
Laundering		
Color Change	3.4	AATCC 61, Test 3A 3/
Staining	3.4	
Sea Water		
Color Change	3.4	AATCC 106, 3/
Staining	3.4	
Light		
Crocking		
Dry	3.4	AATCC 16.2 or 16.3, 4/
Wet	3.4	AATCC 8, 5/

1/ One 6-inch x 6-inch test specimen shall be used to determine compliance. The results shall be recorded as pass or fail.
2/ Shall be rated using AATCC EP1 and AATCC EP2.
3/ Shall be rated using AATCC EP1.
4/ Shall be rated using AATCC EP8.
5/ The test specimens shall be subjected to one (1) washing and drying cycles before the shrinkage and/or elongation values are calculated.

4.4.2 Visual shade matching. The finished cloth shall be evaluated for shade match, appearance, and the presence of optical brightener under ultraviolet source. The finished cloth shall be viewed using AATCC Evaluation - Procedure 9, Option C, under lamps which approximate artificial daylight D75 illuminant having a correlated color temperature of 7500 (± 200) Kelvin, with illumination of 100 (± 20) foot candles to determine if it is a good match to its respective standard sample, see 3.2.4, 3.2.6 and 3.9.

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4.4.3 Visual examination. The cloth shall be examined on face side only for all defects defined in ASTM D3990 and Section 4 which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit. No linear yard (increment of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45 points. If one roll exceeds the point level per 100 square yards, a second sample of 20 rolls shall be examined only for the individual roll quality. If one or more rolls in the second sample of 20 exceeds the individual point value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

4.4.3.1 Demerit point computation.

$$\text{Points per 100 square yards} = \frac{3600\ T}{(W \times Y)}$$

Legend:

T = total number of points scored in the sample
W = width (in inches) of the cloth purchased
Y = total number of yards inspected

4.4.3.2. Demerit points. Demerit points shall be assigned as follows.

a. For defects up to and including 3 inches in any dimension – one point.
b. For defects exceeding 3 inches but not exceeding 6 inches in any dimension – two points
c. For defects exceeding 6 inches but not exceeding 9 inches in any dimension – three points
d. For defects exceeding 9 inches in any dimension – four points.

The following defects, when present, shall be scored four points for each yard in which they occur.

e. Baggy, ridgy or way cloth
f. Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
g. Excessive neppiness

4.4.4 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only as specified in 4.4.1. When the total yardage in the roll does not exceed 100 yards the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards only 100 yards shall be examined. Only one roll shall be taken from any shipping container, unless the number of shipping containers in the lot are less than the number of rolls required for sampling, in which case, all shipping containers shall be sampled.

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4.4.5 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off-shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot presented by the sample, see 3.2.4 and 6.2.

4.4.6 Width examination. During the yard-by-yard examination, each roll in the sample shall be examined for minimum acceptable width (see 3.5) and selvage width when applicable (see 6.2).

4.4.7 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for gross length. When total yardage in the roll exceeds 100 yards, only 100 yards shall be examined, see 3.6 and 6.2.

4.4.8 Put-up examination. During the yard-by-yard examination, each roll in the sample shall be examined for the cloth's face side being inward, see 3.6.1.

4.4.9 Fiber identification label examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance to the 16 CFR Part 303 - Rules and Regulations under the Textile Fiber Product Identification Act. The lot shall be unacceptable if two or more rolls in the sample when the fiber identification is missing from either or both ends, two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Product Identification Act see 3.7 and 6.2.

4.4.10 Face identification marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance to the face identification, 3.8. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample are not labeled (see 3.8 and 6.2).

4.4.11 Total yardage in sample examination. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls, see 3.6 and 6.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended Use. The acrylic bunting cloth covered by this specification is primarily intended for use in the manufacture of signal flags.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number, and date of this specification.
b. National stock number.
c. The specific issue of individual documents reference (see Section 2).
d. When first article is required (see Section 3.1, Section 4.2).
e. When required, yard-by-yard examination (see Section 4).
f. Name and address of the first article inspection facility; and the name and the address of the Government activity responsible for conducting the first article inspection program.
g. Inspection conditions.
h. Conformance inspection and acceptable quality levels (see Section 4.3)
i. Packaging required (see Section 5.1).

6.3 Information requests. For access to information such as patterns, drawings, standard shade samples of cloth etc. go to Defense Supply Center Philadelphia's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military.

6.4 Supersession data. This document supersedes NU PD 2277E dated 23 March 2021.

6.5. Subject term (key word) listing.

Flag fabric
Flag material
Pennant fabric
Pennate material

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy-NU
DLA-CT

Preparing Activity:
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Agent:
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